

Yawar Badri

Systems Engineering | Hardware Architecture | Power Electronics

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Professional Summary

Electrical engineer and systems architect with experience in power electronics, embedded systems, hardware design, and engineering tools for space and terrestrial applications. Work spans system architecture, PCB design, firmware, verification, and commissioning, including satellite power systems, grid-connected converters, energy storage, and test equipment.

Core Competencies

- **Design & analysis:** KiCad, Altium, Cadence, PCB layout, MATLAB/Simulink, LTspice, Qspice, FMECA, WCCA
- **Firmware & software:** C, C++, C#, Python, JavaScript, STM32, ARM Cortex, embedded DSP
- **Verification:** EGSE, hardware-in-the-loop (HIL), oscilloscopes, logic analyzers, LCR, DMM, power quality analysis

Experience

Pixxel Space

Senior Electrical Engineer

2022 – Present

Space / High-Rel Electronics

- Architected end-to-end electrical power systems (EPS) for LEO satellites, from sizing and component selection through assembly. Designed buck power tiles for up to 100 V, achieving approximately 800 W/in³ power density.
- Led small teams through design, test, and commissioning of Firefly satellites, coordinating with thermal, mechanical, AOCS, and AIT teams.
- Built local component search tools and databases to support COTS part selection, reducing procurement lead times from over six months to one to four weeks.
- Developed electrical ground support equipment (EGSE), including a modular 500 W solar string simulator and multichannel power loggers.

Kashmir University

Technical Advisor & Mentor

2021 – 2025

Hardware Startups & Prototyping

- Advise students and early-stage hardware teams on system architecture, prototyping, and embedded integration.
- Mentored PhD students in hardware design and conducted embedded systems workshops.

NIT Srinagar

Project Associate & Mentor

2022 – Present

Power Systems

- Led hardware architecture and systems integration for an experimental 10 kW grid-tied inverter testbed.
- Mentor students in hardware and power electronics.

University of Kashmir (Sponsored by J&K Govt.)

Research Analyst / Associate

2021

EV Infrastructure Analysis

- Analyzed distribution constraints for EV infrastructure and translated technical findings into state policy recommendations.

Selected Hardware Projects

- **10-channel power logger:** Designed a bidirectional DC logger rated up to 100 V, including analog front end, STM32 USB interface, firmware, calibration, and desktop plotting application.
- **DACard-Basic HIL interface:** Built hardware, embedded firmware, host software, and Python libraries for a hardware-in-the-loop test platform.
- **Three-phase grid-tied inverter:** Designed the board layout for power conversion supporting active and reactive power control research.
- **Ground energy storage BMS:** Designed and deployed battery management systems for ground-based energy storage.

Software & Engineering Tools

- **WayriCAD (formerly KiWay):** Open-source KiCad plugin suite for PCB design and analysis, including fanout routing, via stitching, BOM editing, trace RLC, and power and signal integrity checks.
- **CATS:** CAN-based automated hardware testing with a command, read, verify, and record workflow through PCAN hardware.
- **sigLib:** C++ DSP library for real-time waveform conditioning, smoothing, and analysis on embedded targets.
- **Custom engineering applications:** Developed transformer design, data acquisition, load-profile, and billing applications.

Interactive Engineering Utilities

- Built browser-based calculators for SMPS topologies, loop compensation, motor dynamics, PV yield, battery configuration, filters, trace sizing, and magnetics.
- Developed a Web Serial interface for live hardware data monitoring and plotting.

Education

Lovely Professional University (LPU)	2022
<i>Master of Technology (M.Tech) in Electrical Engineering (Power & Renewable Energy)</i>	<i>Grade: A</i>
University of Kashmir	2019
<i>Bachelor of Technology (B.Tech) in Electrical Engineering</i>	<i>Grade: A</i>

Selected Publications

- *Ultracapacitor Integration with Regenerative Braking Systems in Electric Drive Trains* (ScienceDirect/Elsevier, 2023)
- *Experimental Implementation of Optimally Tuned Multifunctional RLMLS Driven VSc Tied to Single Phase Voltage Weak Distribution Grid* (Springer, 2023)
- *Application of V2V Energy Sharing in Electric Vehicles with Source Switching* (IEEE Xplore, 2022)
- *A New Modular Multilevel Converter Topology Using Flying Ultra-capacitor and Cascaded H-bridges* (Springer, 2021)